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FREEDOM Air
Accuracy Test Report

2025.12 | DOF PM 1

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Index

1. Scan Data Accuracy Test
2. PIM Data Accuracy Test

1. Scan Data Accuracy Test

Single case (μm)	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	Overall Avg.
Gom Inspector Deviation	11	13	17	11	13	9	16	12	11	8	12
Quadrant case (μm))	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	Overall Avg.
Gom Inspector Deviation	14	21	13	2	18	23	19	16	19	18	18
Full Arch case (μm)	1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th	Overall Avg.
Gom Inspector Deviation	23	22	18	19	21	17	22	22	27	21	21

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1. Test Procedure

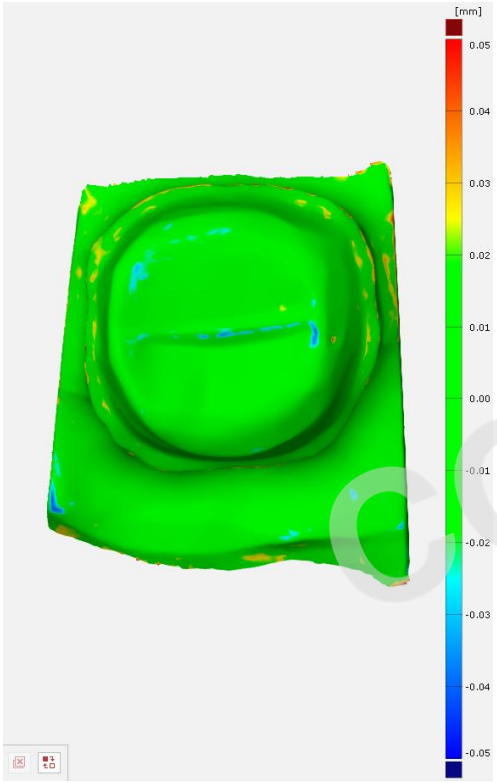
1.1 The test was conducted using a single, a quadrant, and a full-arch stone model.
1.2 Each model was scanned 10 times by the same operator under recommended scanning conditions and strategy.
1.3 Reference data was obtained using our lab scanner FREEDOM X5, and trueness was measured using the Gom Inspector software.
1.4 The average value of the trueness data from the 10 scans was calculated.

2. Conclusion

2.1 The average trueness values were measured as follows:
Single : **12 μm**
Quadrant **18 μm**
Full-arch **21 μm**
2.2 The color maps generated using Gom Inspector have been attached in the following pages.

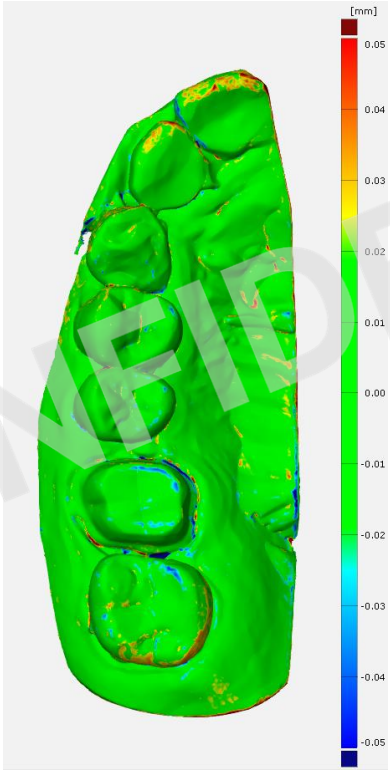
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1. Scan Data Accuracy Test



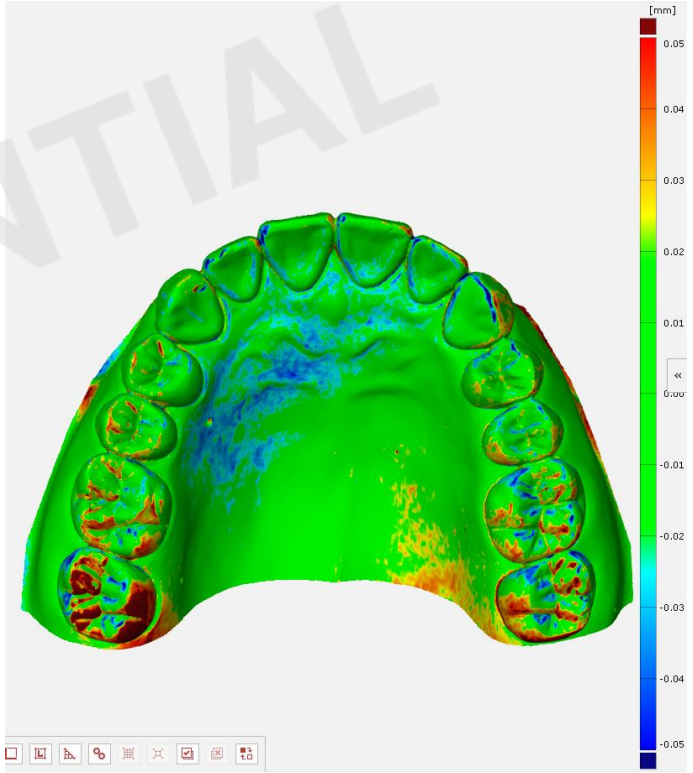
Single

Overall Avg.
12 µm



Quadrant

Overall Avg.
18 µm



Full Arch

Overall Avg.
21 µm

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2. PIM Data Accuracy Test

Distance between scan bodies

6-3 Model		UHD	PIM Aver	Diff	#1	Diff	#2	Diff	#3	Diff	#4	Diff	#5	Diff
SD Linear (mm)	SB1-2	12.025	12.030	**-0.005**	12.029	-0.004	12.028	-0.003	12.032	-0.007	12.026	-0.001	12.036	-0.011
	SB1-3	26.995	26.988	**0.007**	26.979	0.016	26.983	0.012	26.999	-0.004	26.986	0.009	26.995	0.000
	SB1-4	40.241	40.216	**0.025**	40.210	0.031	40.205	0.036	40.216	0.025	40.221	0.020	40.226	0.015
	SB1-5	41.293	41.258	**0.035**	41.258	0.035	41.260	0.033	41.254	0.039	41.255	0.038	41.263	0.030
	SB1-6	42.429	42.420	**0.009**	42.421	0.008	42.422	0.007	42.415	0.014	42.423	0.006	42.419	0.010
	SB2-3	15.404	15.399	**0.005**	15.390	0.014	15.394	0.010	15.406	-0.002	15.401	0.003	15.402	0.002
	SB2-4	31.626	31.620	**0.006**	31.615	0.011	31.609	0.017	31.613	0.013	31.631	-0.005	31.634	-0.008
	SB2-5	37.823	37.819	**0.004**	37.817	0.006	37.819	0.004	37.806	0.017	37.822	0.001	37.831	-0.008
	SB2-6	42.058	42.087	**-0.029**	42.088	-0.030	42.084	-0.026	42.073	-0.015	42.094	-0.036	42.095	-0.037
	SB3-4	20.194	20.190	**0.004**	20.190	0.004	20.188	0.006	20.175	0.019	20.199	-0.005	20.200	-0.006
	SB3-5	33.590	33.582	**0.008**	33.578	0.012	33.590	0.000	33.567	0.023	33.586	0.004	33.590	0.000
	SB3-6	41.338	41.362	**-0.024**	41.362	-0.024	41.363	-0.025	41.349	-0.011	41.369	-0.031	41.367	-0.029
	SB4-5	19.393	19.384	**0.009**	19.378	0.015	19.385	0.008	19.382	0.011	19.389	0.004	19.386	0.007
	SB4-6	29.887	29.901	**-0.014**	29.903	-0.016	29.894	-0.007	29.899	-0.012	29.908	-0.021	29.903	-0.016
	SB5-6	11.099	11.116	**-0.017**	11.125	-0.026	11.109	-0.010	11.117	-0.018	11.115	-0.016	11.116	-0.017

Angle between scan bodies

6-3 Model		UHD	PIM Aver	Diff	#1	Diff	#2	Diff	#3	Diff	#4	Diff	#5	Diff
SD Angular (Deg)	SB1-2	3.26	3.13	**0.13**	3.12	0.14	3.10	0.16	3.09	0.17	3.16	0.10	3.20	0.06
	SB1-3	0.57	0.62	**-0.05**	0.67	-0.10	0.62	-0.05	0.68	-0.11	0.58	-0.01	0.55	0.02
	SB1-4	6.89	7.15	**-0.26**	7.18	-0.29	7.20	-0.31	7.23	-0.34	7.06	-0.17	7.10	-0.21
	SB1-5	3.23	3.50	**-0.27**	3.51	-0.28	3.49	-0.26	3.50	-0.27	3.52	-0.29	3.50	-0.27
	SB1-6	4.24	4.10	**0.14**	4.11	0.13	4.11	0.13	4.09	0.15	4.12	0.12	4.08	0.16
	SB2-3	3.83	3.71	**0.12**	3.74	0.09	3.68	0.15	3.71	0.12	3.69	0.14	3.72	0.11
	SB2-4	8.23	8.23	**0.00**	8.28	-0.05	8.29	-0.06	8.24	-0.01	8.13	0.10	8.23	0.00
	SB2-5	2.00	2.04	**-0.04**	2.07	-0.07	2.04	-0.04	1.97	0.03	2.05	-0.05	2.06	-0.06
	SB2-6	6.41	6.30	**0.11**	6.31	0.10	6.26	0.15	6.27	0.14	6.33	0.08	6.32	0.09
	SB3-4	6.77	6.83	**-0.06**	6.82	-0.05	6.87	-0.10	6.87	-0.10	6.77	0.00	6.82	-0.05
	SB3-5	3.69	3.85	**-0.16**	3.88	-0.19	3.84	-0.15	3.88	-0.19	3.85	-0.16	3.81	-0.12
	SB3-6	4.03	4.09	**-0.06**	4.10	-0.07	4.10	-0.07	4.09	-0.06	4.09	-0.06	4.06	-0.03
	SB4-5	6.33	6.42	**-0.09**	6.44	-0.11	6.48	-0.15	6.49	-0.16	6.30	0.03	6.37	-0.04
	SB4-6	10.52	10.65	**-0.13**	10.66	-0.14	10.71	-0.19	10.71	-0.19	10.59	-0.07	10.59	-0.07
	SB5-6	7.26	7.44	**-0.18**	7.47	-0.21	7.43	-0.17	7.41	-0.15	7.47	-0.21	7.41	-0.15

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1. Test Procedure

1. 1 Fasten the PIM scan bodies onto a single stone model according to the numbers of lab analog and perform 5 repeated scans.

1.2 During the 5 scans, no positional changes of the attached scan bodies.

1.3 Record the linear and angular differences between the two scan bodies for each test run, then calculate the average.

1.4 Using FREEDOM UHD, obtain the reference data and measure the distance and angle between each scan bodies.

1.5 Compare the numerical values (means and per-iteration values) of the PIM data with the reference data. The resulting values correspond to the table above.

1.6 The acceptable tolerance ranges for linear error (mean ± SD linear discrepancies) and angular error (mean ± SD angular discrepancies) are **±30 µm and ±0.34°, respectively**.

2. Conclusion

2.1 When converting the linear error in the above table to micrometers, the Distance average result is **13.4 µm,** the Angle average result is **0.12°**

2.2 All angular errors are confirmed to fall within the acceptable tolerance range. The acceptable tolerance was established based on the findings reported in the attached study.

2.3 *Attached Study*: Accuracy of complete arch implant scans recorded by using intraoral and extraoral photogrammetry systems.

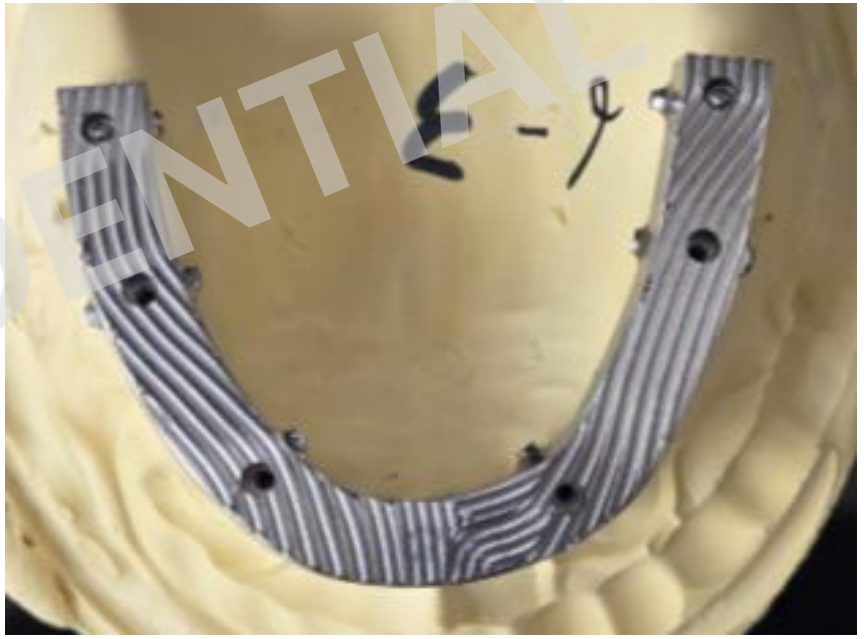
2.4 Using the obtained data, metal bar milling was fabricated with DOF CRAFT PLUS, and the trial fitting photos have been attached in the following pages

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2. PIM Data Accuracy Test



All on 6 Plaster Test Model



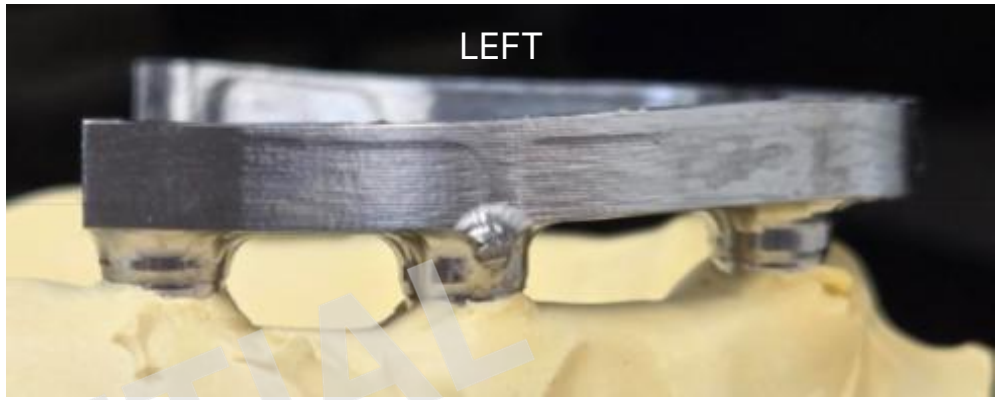
All on 6 Metal Milled Bar

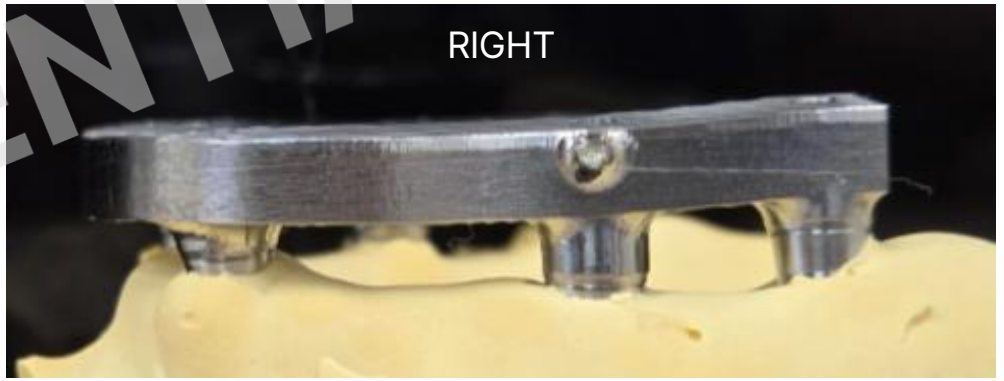
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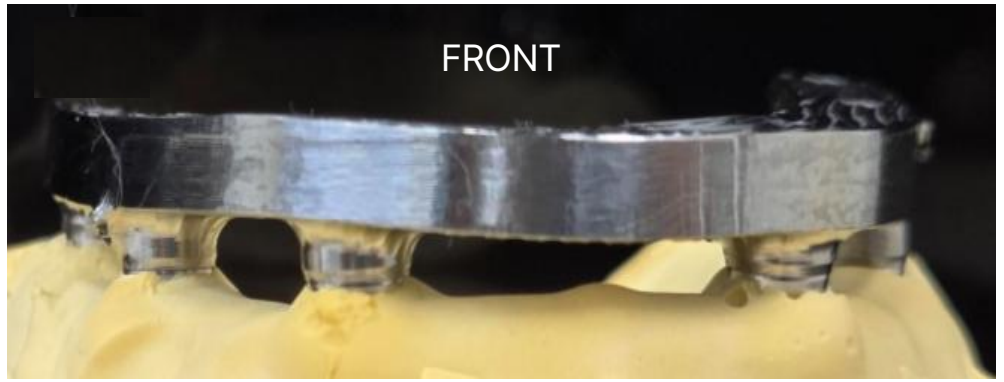
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2. PIM Data Accuracy Test

Trial Fitting Standard	Results
The entire bar structure seated perfectly on all implants without any distortion	PASS
The screw holes aligned precisely with the implants, allowing smooth insertion	PASS
When a screw was tightened on one side, the opposite side did not lift or gap	PASS







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